YASHAS CHANDRA BATHINI *AI Engineer | LLM | RAG | Agentic AI | Python | FastAPI | React | Azure*

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yashas144.github.io/Yashas-Portfolio F-1 OPT | Authorized to Work in the U.S.

PROFILE

AI Engineer with 5 years of experience designing and deploying Agentic AI systems, LLM-powered applications, MLOps pipelines, distributed microservices, and cloud-native platforms. Experienced in autonomous AI agents, Retrieval-Augmented Generation (RAG), real-time analytics, and scalable API architectures. Proven expertise building production-grade AI solutions using Python, Java, FastAPI, React and Azure. Strong background in event-driven systems, cloud deployments, and high-performance AI apps serving mission-critical business operations.

SKILLS

Agentic AI & LLMs: RAG, Agentic AI, LangChain, MCP, Crew AI, Transformers, Hugging Face, NLP, FAISS, ChromaDB.

Full-Stack Web Development: FastAPI, React, REST APIs, Microservices, Event-Driven Architecture.

MLOps: MLflow, Experiment Tracking, Model Registry/Monitoring/Versioning, AI Evaluation, Pydantic Validation.

Cloud & DevOps: Azure, GCP, Docker, Terraform, Git, GitHub Actions, CI/CD, Containerization.

Programming: Python, Java, OOP, Design Patterns, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Cursor, Copilot.

Data Engineering: Databricks, ETL Data Pipelines, Kafka, Tableau, MySQL, PostgreSQL.

WORK EXPERIENCE

AI Data Engineer 10/2024 – Present
Brillfy Technology Inc Irving, Texas, US

AfterHours AI - AI Voice Booking & Agent Orchestration Platform

- Designed and deployed a production-grade conversational AI orchestration system, integrating an AI voice agent with FastAPI services and external APIs to automate appointment workflows with zero manual data entry.
- Engineered asynchronous, event-driven processing and WebSocket communication, supporting 10,000+ daily events with sub-second real-time updates to a React dashboard.
- Built resilient distributed workflow controls including event-ID idempotency, retry-safe API integrations, and duplicate-event prevention across Google Calendar and Sheets APIs.

AI-Powered Healthcare Payer & Prior Authorization Platform

- Developed an AI-powered healthcare payer platform supporting Eligibility, Benefits, Provider Management, Claims, and Prior Authorization workflows using Python, FastAPI, React, REST APIs, and microservices.
- Built an AI Prior Authorization Copilot using Azure OpenAI, RAG, vector search, and agentic workflows to analyze clinical documents, retrieve payer policies, identify missing information, and provide decision support.
- Implemented FHIR APIs and simulated HIPAA X12 270/271 eligibility and 278 prior-authorization workflows using synthetic healthcare data.
- Engineered cloud-native services with OAuth 2.0/OIDC, JWT, Docker, Azure Service Bus, Terraform, GitHub Actions CI/CD, automated testing, and Application Insights.

AI Full-Stack Engineer 06/2021 – 07/2024
Sysarket Hyderabad, India

Auravance – Industrial Predictive Intelligence Platform

- Built a full-stack AI platform (React, FastAPI) for real-time equipment monitoring and failure prediction using XGBoost and LSTM, improving detection accuracy by 30%.
- Engineered scalable event-driven IoT ingestion pipelines using MQTT and Python for high-frequency sensor telemetry with low latency and high reliability.
- Designed scalable ETL and event-driven MQTT pipelines processing 1M+ sensor records/day, enabling predictive maintenance, anomaly detection, and RUL insights with sub-minute latency.
- Built a RAG-based root cause analysis system over equipment manuals to generate explainable diagnostics and maintenance recommendations.

AI-Powered Music Intelligence Platform

- Designed an AI recommendation platform using **Hybrid RAG (BM25 + vector embeddings)** across **1M+ songs** for context-aware retrieval based on mood, genre, and user intent.
- Architected scalable retrieval pipelines with embedding generation, vector search, hybrid retrieval, and reranking; deployed the full-stack system on **Azure and Hugging Face** with CI/CD.
- Engineered real-time AI features, including real-time mood detection and seamless playback.

EDUCATION

Master of Science in Data Science Denton, Texas
University of North Texas, CGPA: 4.0 / 4.0

Bachelor of Technology in Information Technology Hyderabad, India
Jawaharlal Nehru Technological University Hyderabad, CGPA: 3.7 / 4

CERTIFICATIONS

AWS GenAI Practitioner | Databricks GenAI Fundamentals | UCSD ML with Big Data